Pricing American Options under Regime Switching Volatility using Least Squares Monte Carlo

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Paper to be reproduced

Title: Valuing American Options by Simulation: A Simple Least-Squares Approach

Authors: Francis A. Longstaff, Eduardo S. Schwartz

Journal: The Review of Financial Studies, Vol. 14, No. 1. (Spring, 2001), pp. 113-147.

Stable URL: https://www.jstor.org/stable/2696758

Problem Statement

- The standard Geometric Brownian Motion (GBM) model assumes that volatility is constant over the life of the option. In practice, financial markets exhibit periods of low volatility (calm markets) and high volatility (turbulent markets), and these regimes tend to persist and switch over time.
- This project extends the Longstaff-Schwartz Least-Squares Monte Carlo (LSMC) framework for pricing American put options by replacing the constant volatility assumption with a two-state Markov regime-switching volatility model, where:

  - sigma_t = sigma_1 (low volatility regime, e.g. 0.15)
  - sigma_t = sigma_2 (high volatility regime, e.g. 0.40)

- The goal is to understand how regime-switching volatility affects American option prices and early-exercise decisions

Dataset

- The dataset will be synthetically generated via Monte Carlo simulation, similar to Longstaff & Schwartz (2001).
- Simulation parameters:
  - Initial stock price S0 = 36
  - Strike price K = 40 (in-the-money put)
  - Risk-free rate r = 0.06
  - Time to maturity T = 1 year, with 50 time steps
  - 100,000 simulated price paths
- Regime-switching parameters:
  - sigma_1 = 0.15 (low vol regime), sigma_2 = 0.40 (high vol regime)
  - Transition probabilities p_12 and p_21 governing regime switches
  - Regimes evolve as a 2-state Markov chain at each time step
- No external data is needed. All price paths and regime sequences are generated programmatically in Python using NumPy.

Motivation

- Real financial markets show volatility clustering - long calm periods followed by sudden turbulence (e.g. 2008 crisis, COVID crash). Constant volatility models cannot capture this.
- Regime-switching models are widely used in financial econometrics to describe this two-state behavior (Hamilton, 1989).
- American options are harder to price than European options because of the early exercise feature. LSMC is one of the most popular simulation-based methods for this.
- Combining regime-switching dynamics with LSMC is a natural extension that has practical relevance - option prices depend on the current volatility regime, and ignoring this can lead to mispricing.

Related Papers

Title: American Options with Regime Switching

Authors: Buffington, J. & Elliott, R.J. (2002)

Journal: International Journal of Theoretical and Applied Finance, Vol. 5, No. 5, pp. 497–514.

Summary:

- Derives analytical pricing results for American options when the underlying asset follows a GBM with Markov regime-switching volatility
- The volatility switches between two states governed by a continuous-time Markov chain
- Provides closed-form and semi-analytical solutions that can serve as benchmarks

Why I chose it: We reproduce / validate their pricing results using the Longstaff-Schwartz LSMC simulation method.

Summary of the Method

- Step 1: Simulate regime-switching GBM paths
    - At each time step, sample the regime from a Markov chain
    - Use the regime dependent sigma (volatility) to evolve the stock price
- Step 2: LSMC backward induction
    - Work backward from maturity, comparing immediate exercise value against the estimated continuation value
    - The state is now two-dimensional: (stock price, current regime). The continuation value regression uses both as features.
- Step 3: Compare against constant-vol baseline
    - Price the same option under constant volatility (standard GBM) and under regime switching to quantify the pricing difference
    - Vary the regime parameters (sigma_1, sigma_2, transition probabilities) and observe how the option price changes

What I hope to learn

- How much the option price changes when volatility is allowed to switch between regimes instead of being held constant
- Whether ignoring regime switching leads to systematic over or under pricing of American options
- How the early exercise decision differs across high-vol and low-vol regimes
- Practical experience implementing a regime switching model and connecting it to a simulation based pricing framework
- A deeper understanding of how conditional volatility models from financial econometrics interact with option pricing theory